Overview of Helios Applications at Sikorsky

17th Symposium on Overset Composite Grids and Solution Technology
Sep. 2026

Byung-Young “B.Y.” Min, Jinggen Zhao, Brian Wake

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Outline

- ❑Analysis Hierarchy Schematic
- ❑Applications
  - ❑Blade Design: HSHER
  - ❑Flight Test Support: CH-53K® helicopter
  - ❑Flight Test Support (Loads Prediction): S-97 Raider® aircraft
  - ❑Virtual Flight Before First Flight: Dragonfly
- ❑Validations & Improvements
  - ❑PSU Hub Flow Validation
  - ❑Hovering Rotor: HVAB & XV-15
  - ❑Black Hawk® helicopter Hover & Maneuver
  - ❑Co-Simulation With Flight Control Laws (S-92® aircraft)
- ❑Flight Science Study
- ❑Applications in Development
- ❑Concluding Remarks

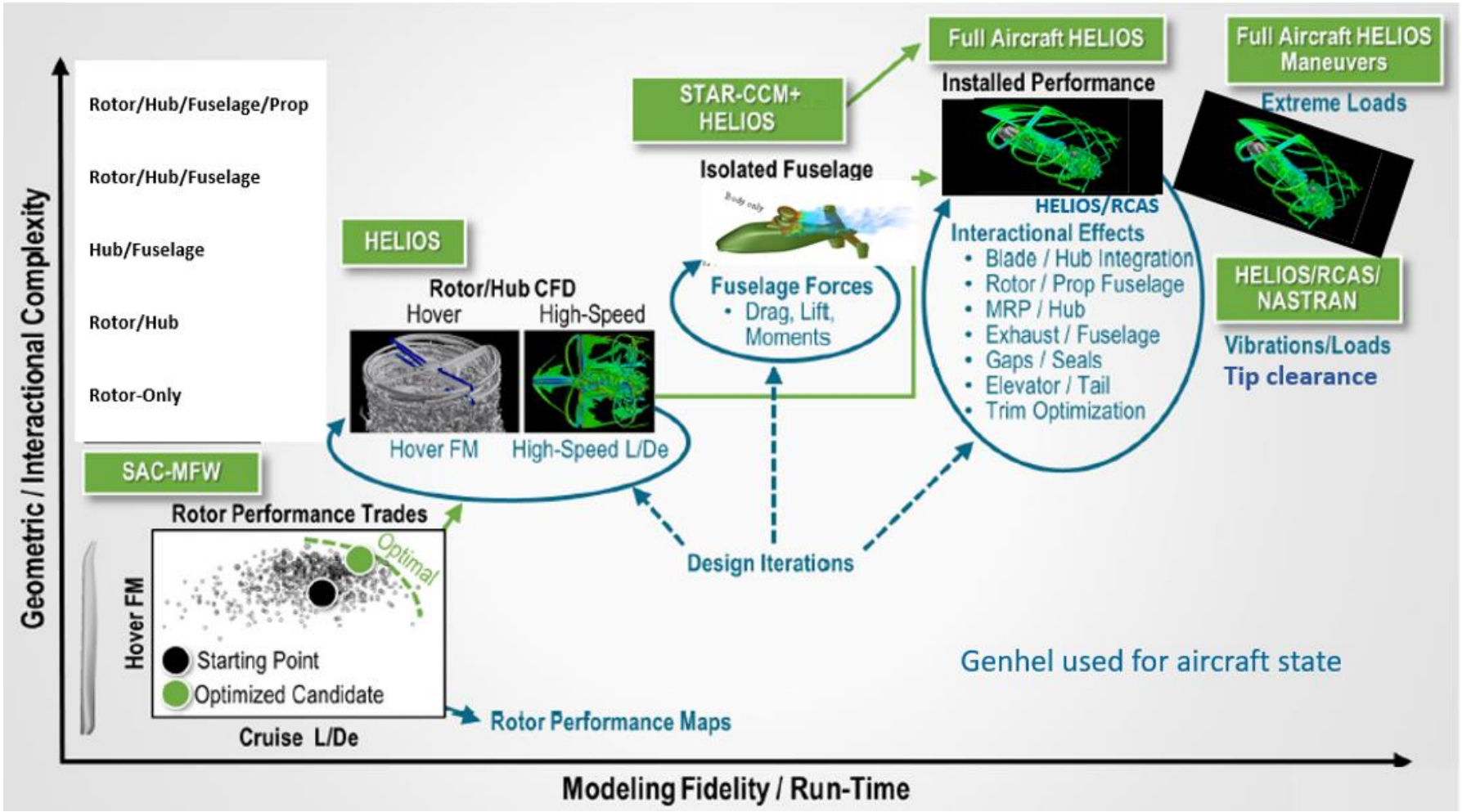
*CH-53K® used with permission of the Department of the Navy

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Analysis Hierarchy During Development

Computational resources and tools enable higher resolution and more evaluations



SIK2023070227

Ref: Active Flow Control for Rotorcraft, Peter F. Lorber, Wake, B.E., Bowles, P..O., and Matalanis, C.G., Active Flow Control Workshop, University of Maryland, July 25-26, 2023.

Est. CPU-days/case:	0.1	200 -1000	5000	20000

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Applications

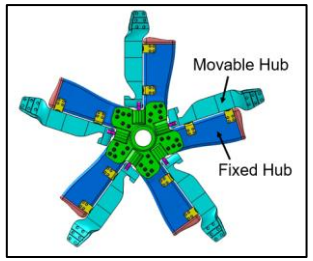
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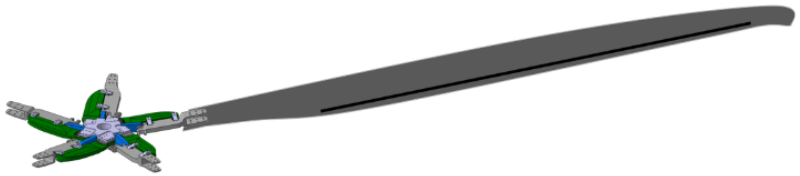
Blade Design Example: HSHER Rotor CFD/CSD

References:
Min, B.-Y., et al, “HSHER High Speed and Highly Efficient Rotor Blade Design,” VFS, May 2022.
Sutton, C.M., et al, “Aeroelastic Design of a High-Speed Highly Efficient Rotor,” VFS, May 2023.

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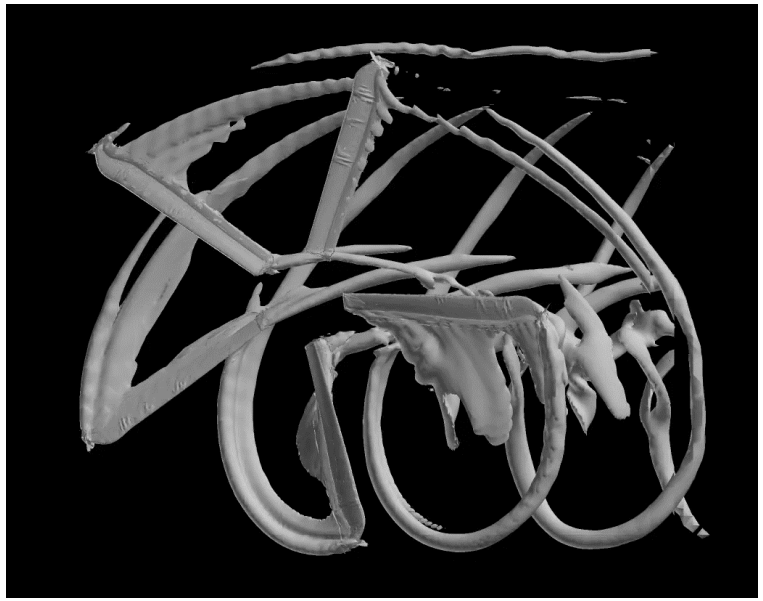
HSHER Rotor Calculations for Performance and Vibrations

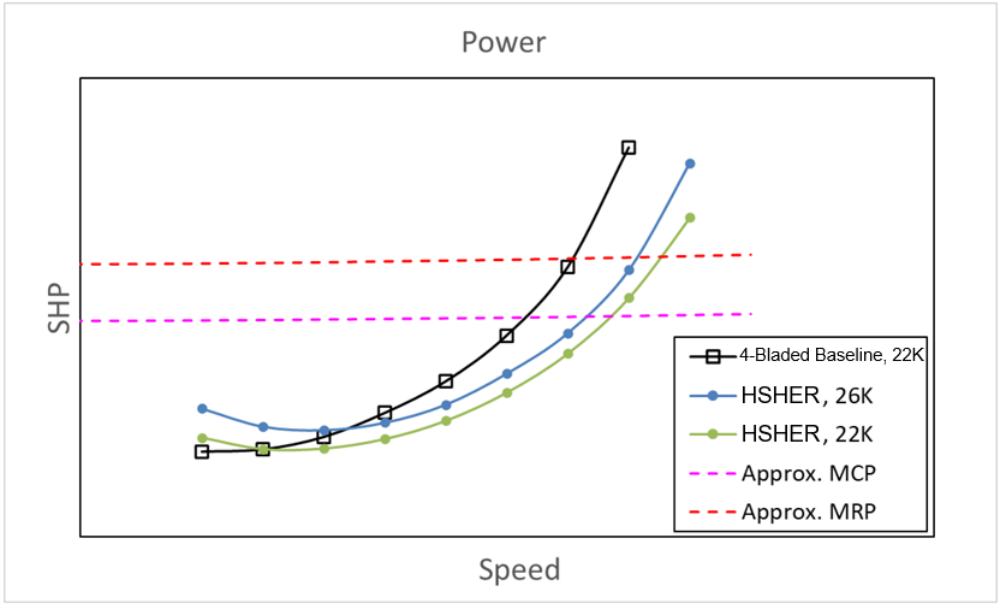




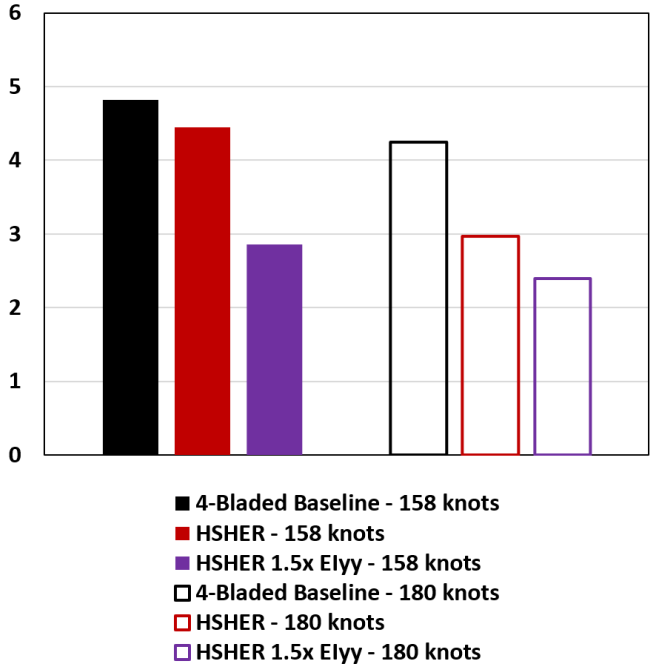


HSHER at 180 knots





Vibrations Analysis



HSHER: High-Speed Highly-Efficient Rotor

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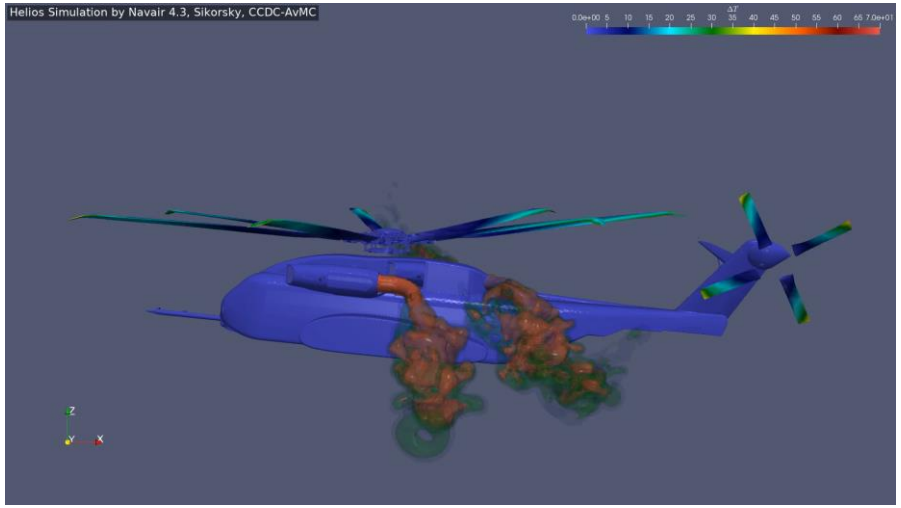
Flight Test Support: CH-53K® King Stallion

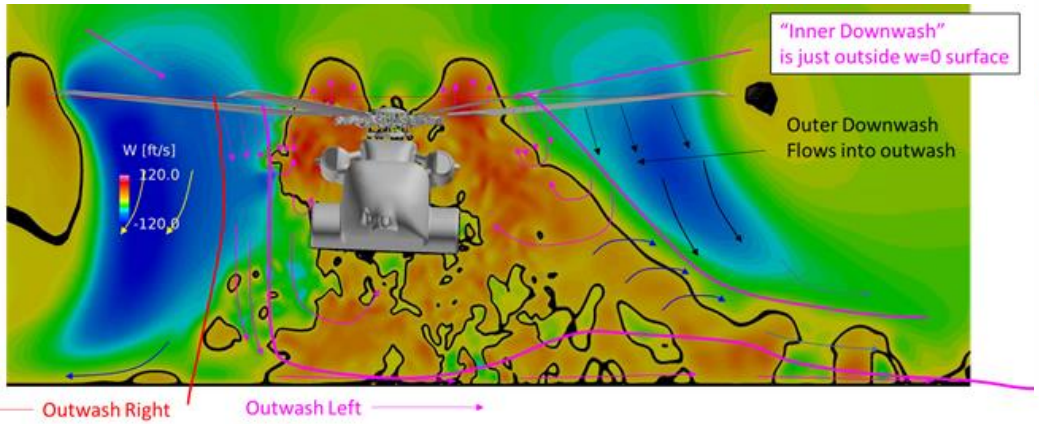
References:

SIK202010021 Min, B. Y., et al, "Study of Exhaust Gas Reingestion," AIAA Sci Tech Conference, Jan 2021.

SIK2023040155 Klimchenko, V., et al, "Full-Aircraft CH-53K® Hover Simulations with Helios," 79th VFS Forum, May 2023.

Exhaust Gas Reingestion (EGR)

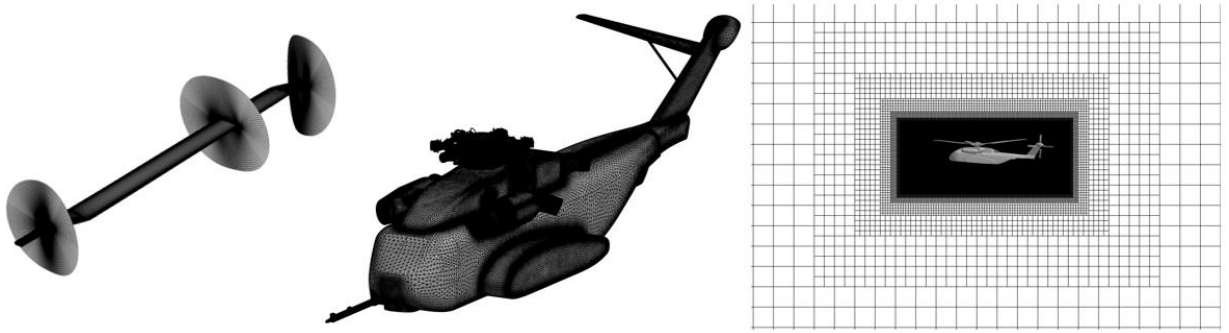




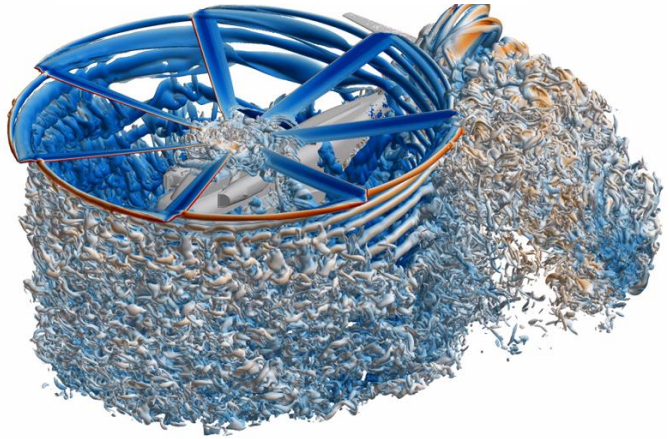
EGR examined and resolved with CFD
VFS Hughes Award - 2022

Full-Aircraft Hover Performance

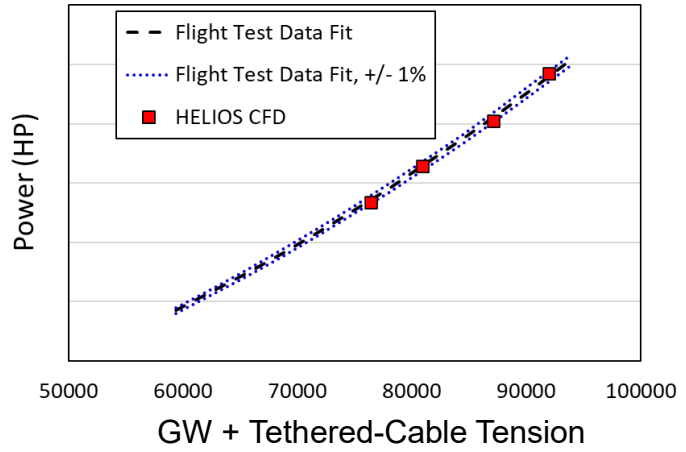
HELIOS Grid System for CH-53K®



Vorticity Iso-Surfaces in Hover



CH-53K® Power in Tethered-Hover Test



CH-53K® used with permission of the Department of the Navy

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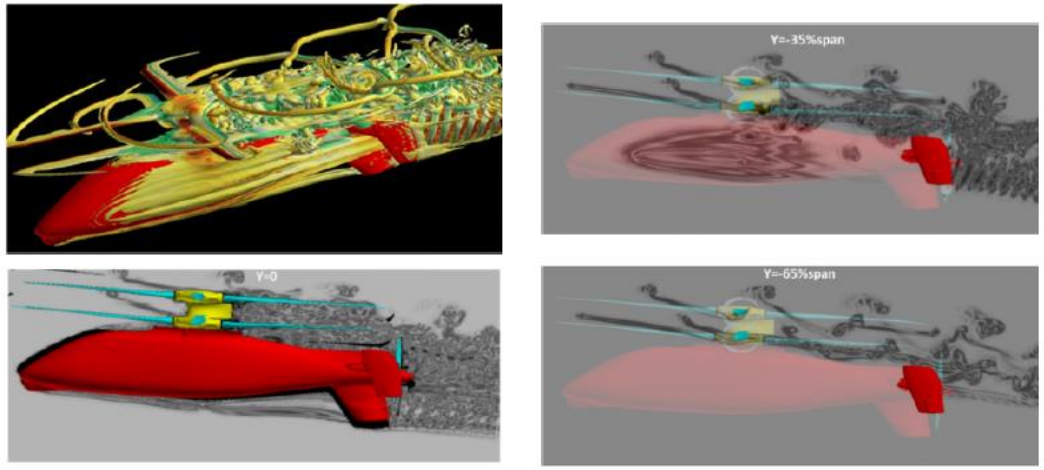
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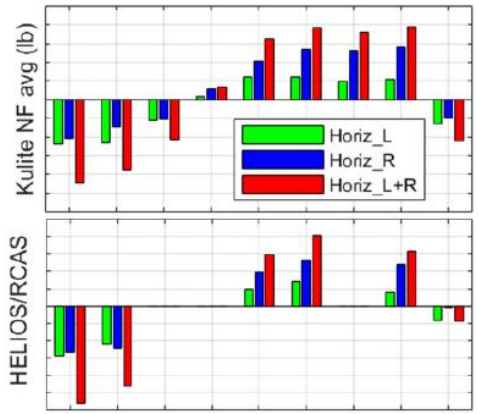
References:
Lorber, P.F., Zhao, J., et al, “S-97 Raider® Wake-Empennage Interaction Flight Data and Correlation,” VFS Forum, May 2021.
Zhao, J., et al., “S-97 RAIDER® Maneuvering Loads: Flight Test and Simulation Correlation,” VFS Forum, May 2023.

Flight Test Support (Loads Prediction): S-97 Raider®

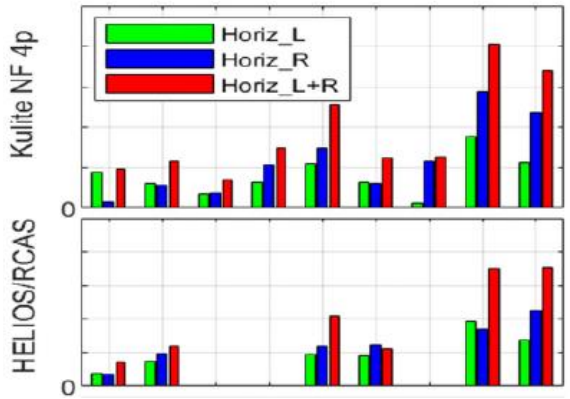
Tail Loads in Level Flight



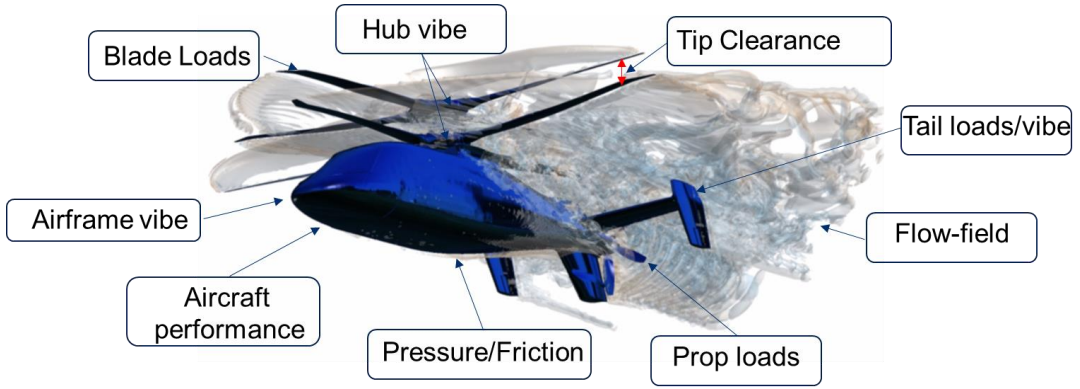
Steady Tail Loads



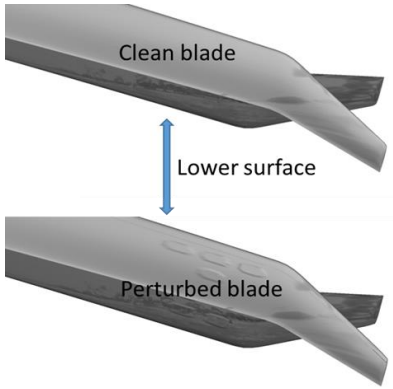
4P Tail Loads

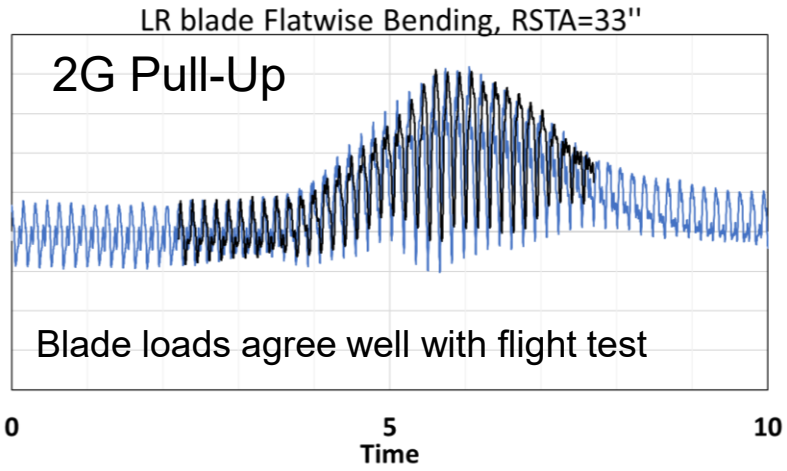


Maneuvering Flight Blade Loads



Blade Imperfections Included in Latest Models





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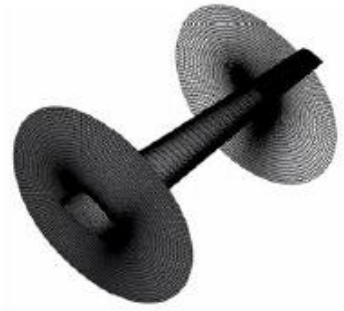
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Virtual Flight Before First Flight: Dragonfly

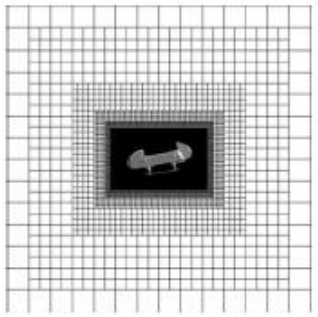
Dragonfly Calculations for Performance and Vibrations



KCFD (Lander/Arm/Hub)

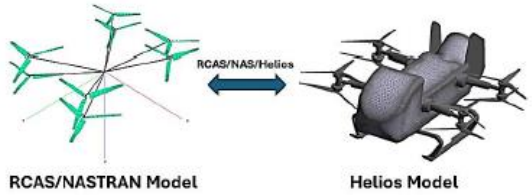


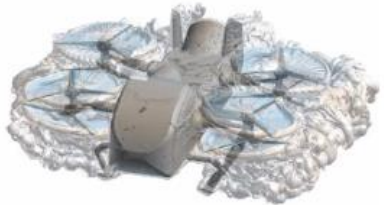
OVERFLOW (Rotor blades)



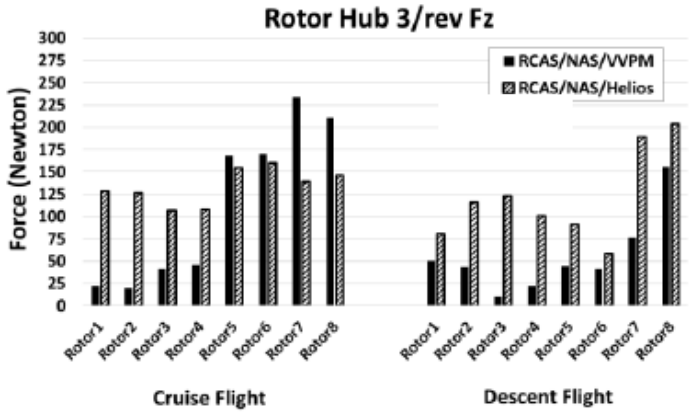
SAMCART (Off-body)

Helios Grid Systems





Coupled HELOS/RCAS/NASTRAN Simulations

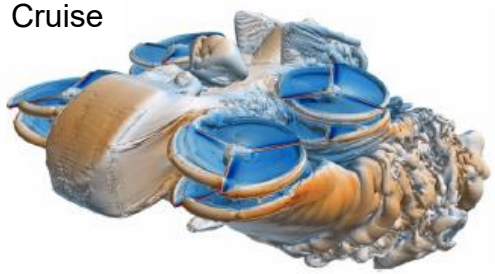


References:
Vermillion, T.-A., et al, “Dragonfly Hub Loads Correlation with Wind Tunnel Data,” VFS 81st Forum, May 2025.
Modarres, R., et al, “Low-Vibration Design of the Dragonfly Lander,” VFS 81st Forum, May 2025.

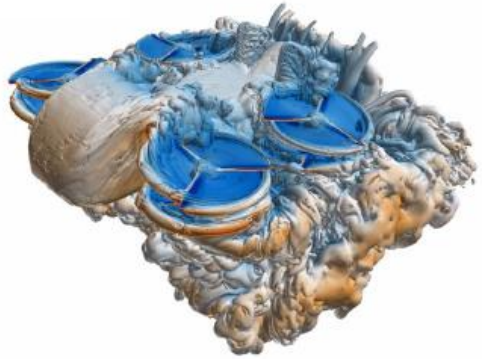
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Cruise



Descent



Flowfield in Cruise and Descent Conditions

Rotorcraft for Saturn’s Moon (Titan), NASA Program

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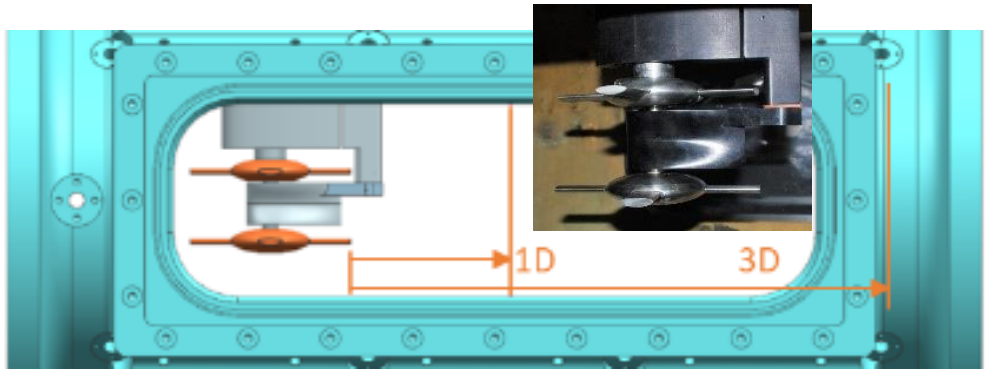
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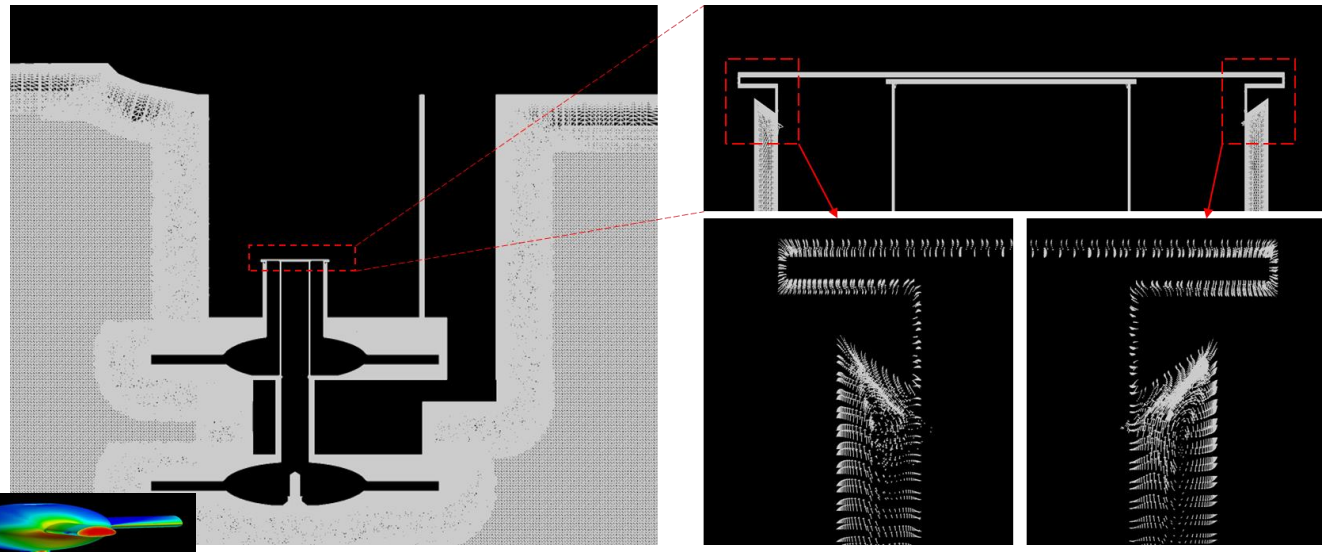
Validations & Improvements

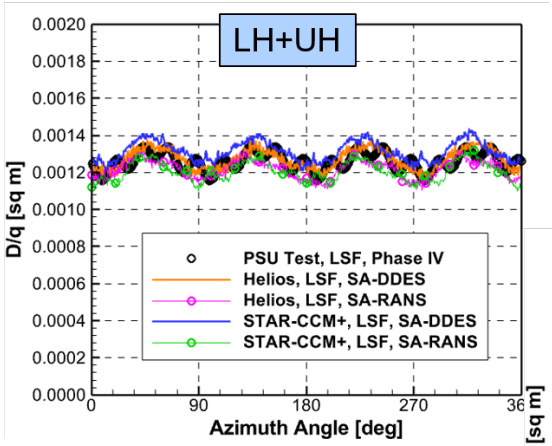
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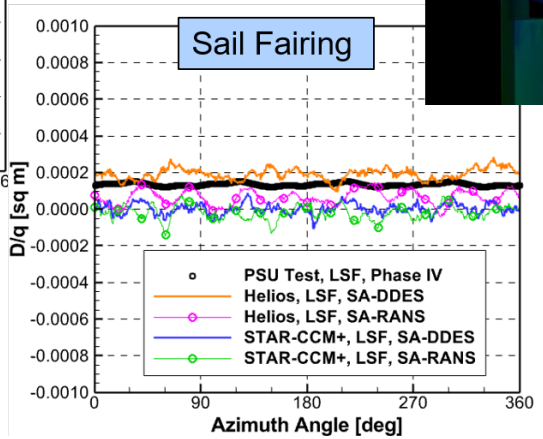
PSU Hub Flow Validation

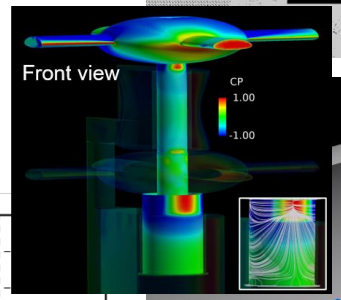
Gap Treatment

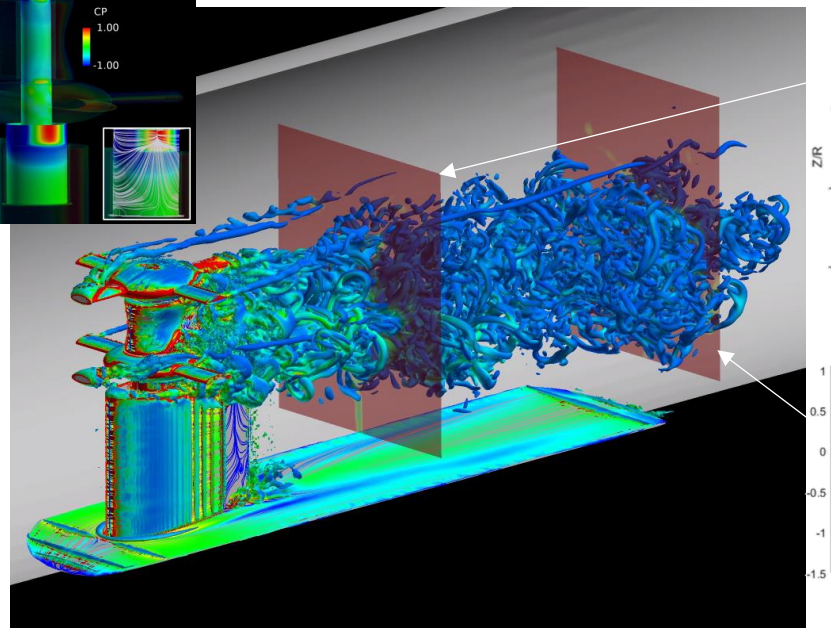


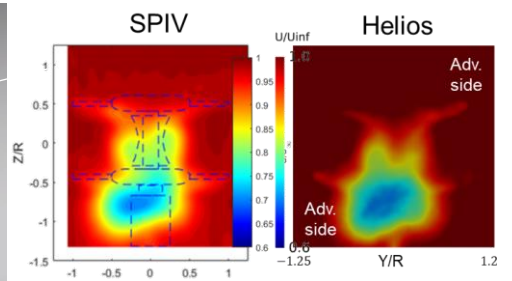


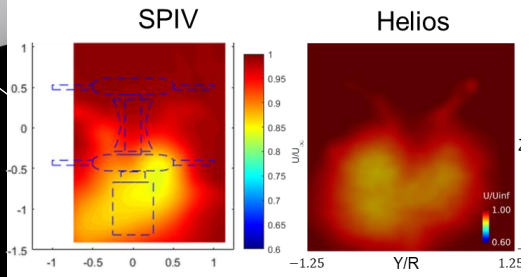












SIK202208004

References:
Min, B.-Y., et al, “Coaxial Hub Drag Correlation with a Water-tunnel Model using Two Flow Solvers,” VFS 79th Forum, May 16-18, 2023..

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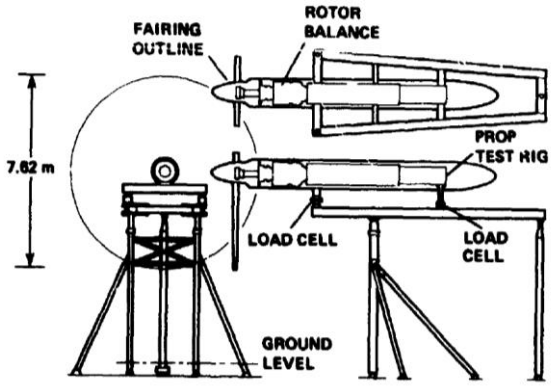
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Hovering Rotor: HVAB & XV-15 Correlation

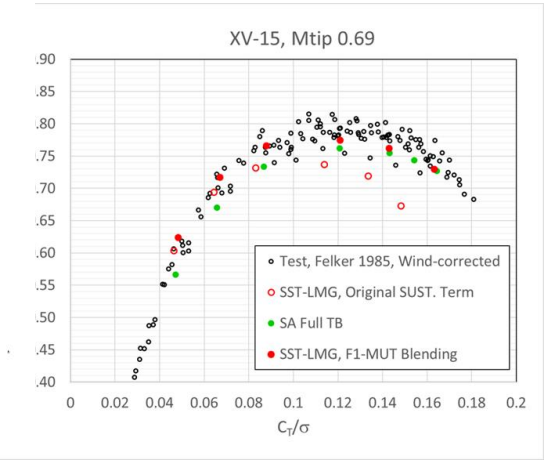
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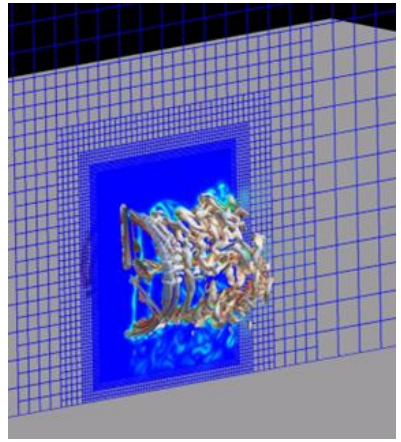
References
Min, B.-Y., Jain, R., and Wake, B.E., “An Investigation of CFD Predictions for Rotor Hover Performance Near Stall Conditions,” AIAA Sci. Tech. Conference, Jan. 2026.
Min, B.-Y. and Wake, B. E., “Turbulent Transition Model Validation with Multiple Rotors for Industry Application,” AIAA Aviation Forum, July 2025.

Validation and model improvements with HVAB & XV-15 data useful for existing and new programs



XV-15 Test Rig

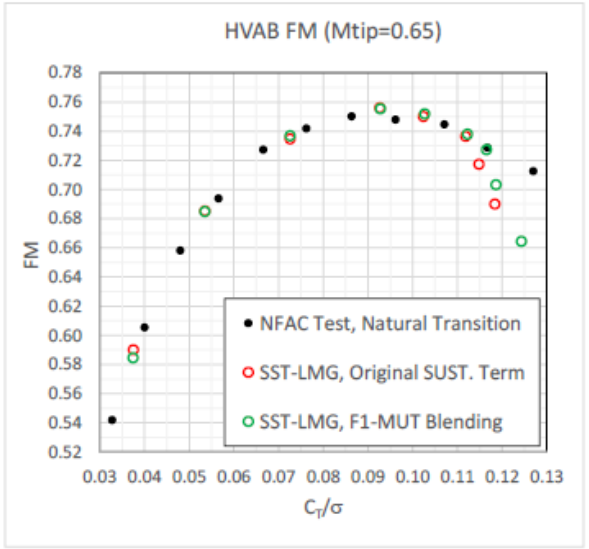


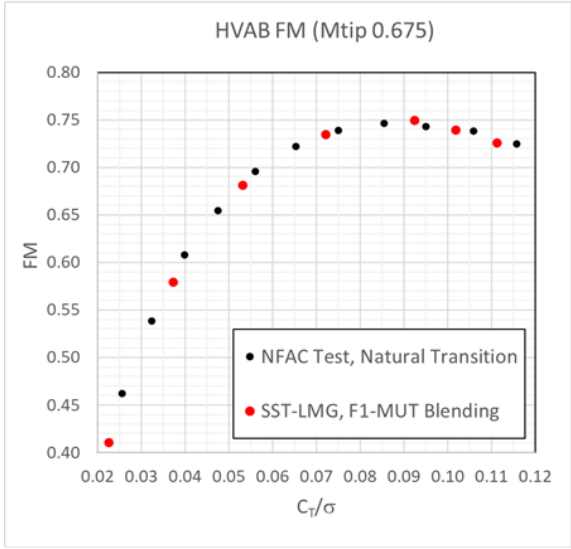


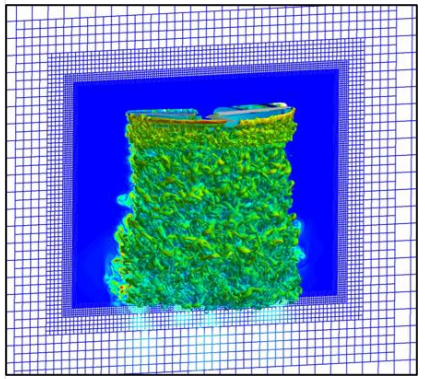


HVAB Rotor Installed in NFAC

HVAB – Hover Validation and Acoustic Baseline







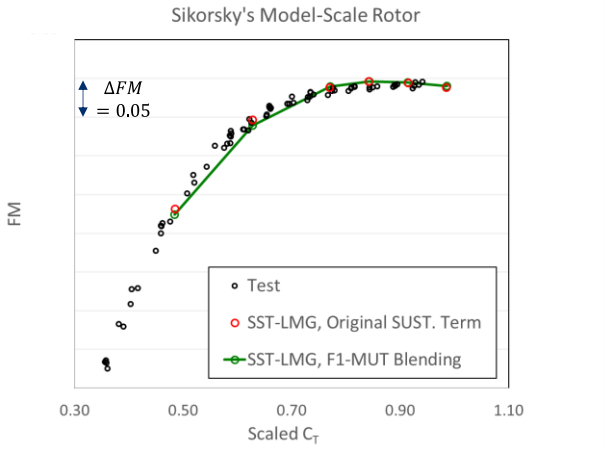
Transition model and modifications to turbulence model improve correlation in hover

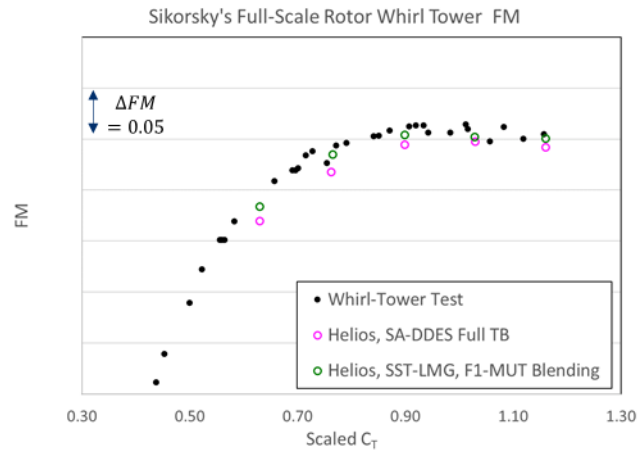
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Recent Examples of CFD Capabilities: Black Hawk® Helicopter

Black Hawk® (BH) Calculations for Performance and Unsteady Structural Loads





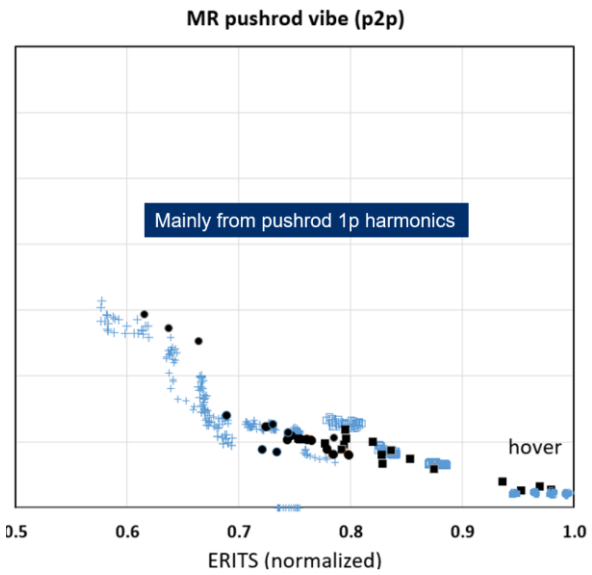


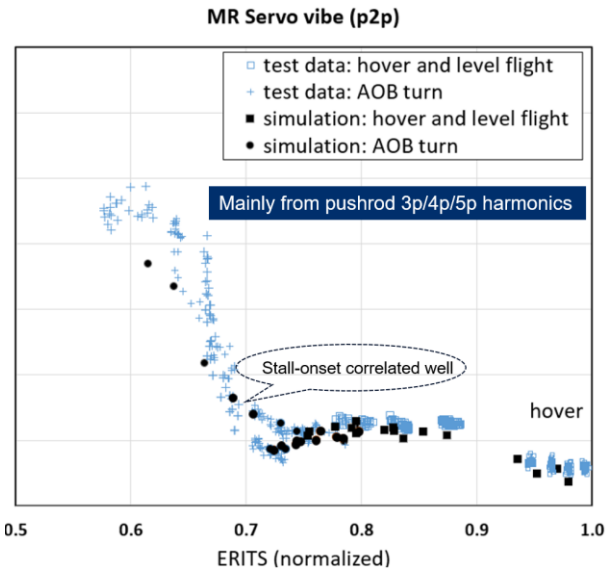
Sikorsky model-scale rotor



Sikorsky full-scale rotor

Black Hawk® Control Loads Correlation (level flight & AOBs)





Correlation for Black Hawk® Modernization Program

References:

Min, B.-Y. and Wake, B. E., "Turbulent Transition Model Validation with Multiple Rotors for Industry Application," AIAA Aviation Forum, July 2025. SIK2025050781

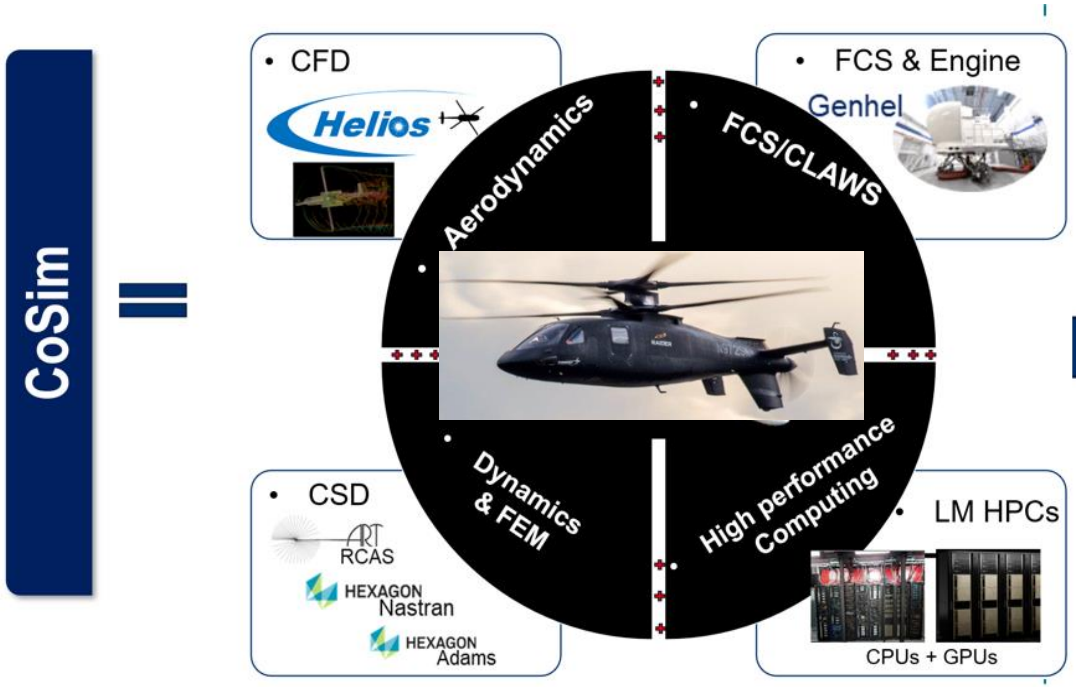
Zhao, J., et. al., "Full-Aircraft UH-60M Loads and Performance Correlations at Level and Maneuvering Flight Conditions with HELIOS/RCAS," VFS 81st Forum, May 2025. SIK2025030699

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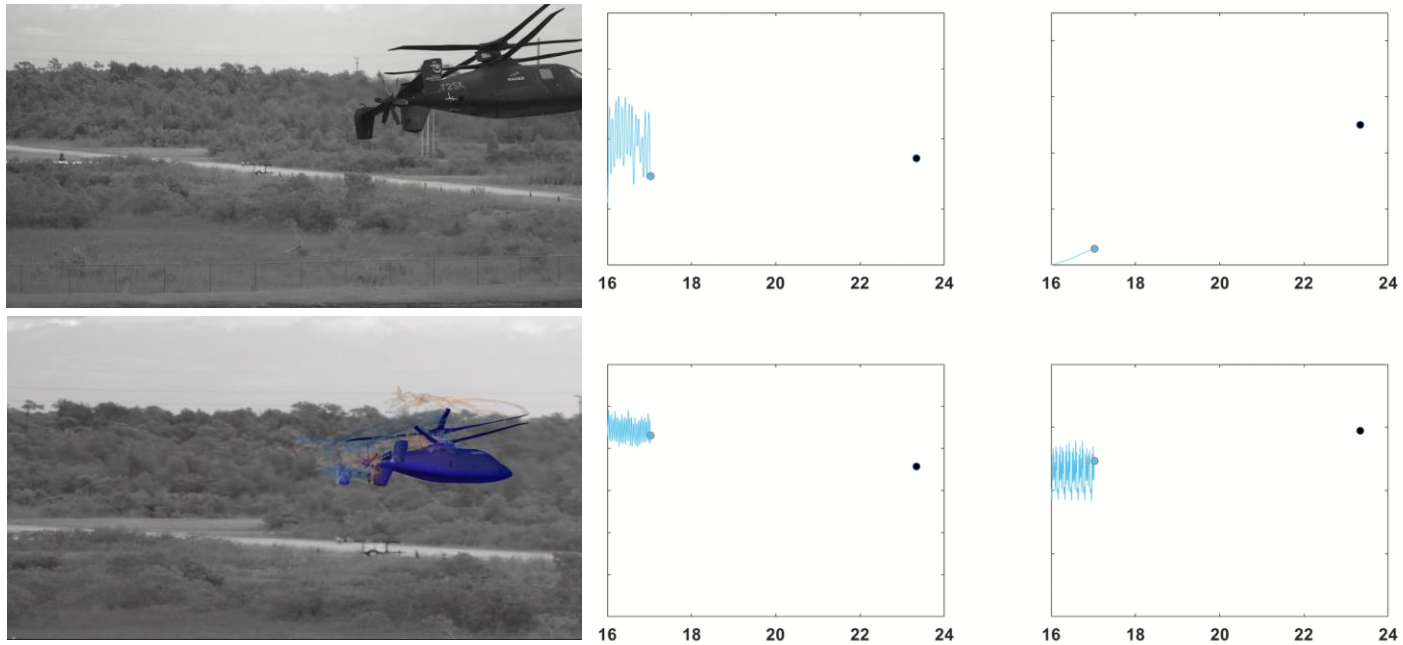
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Integration with Flight Control Laws (FCS): Co-Simulation

Helios/RCAS/FCS Co-Simulation



S-97 Raider® Angle-of-Bank (AOB) Turn Maneuver Correlation



Coupling of flight control laws to CFD/CSD removes dependency of simplified aerodynamics for flight dynamics
VFS Hughes Award - 2024

References:
Zhao, J., et al, “CFD-CSD-FCS Free-Flight Co-Simulation Correlations with S-92A Maneuvering Flight Tests,” VFS 82nd Forum, May 2026.

Zhao, J., et al, “Development and Validation of Full-Aircraft Free-Flight Maneuvering Simulation using Tightly Coupled CFD-CSD and FCS,” VFS 80th Forum, May 2024.

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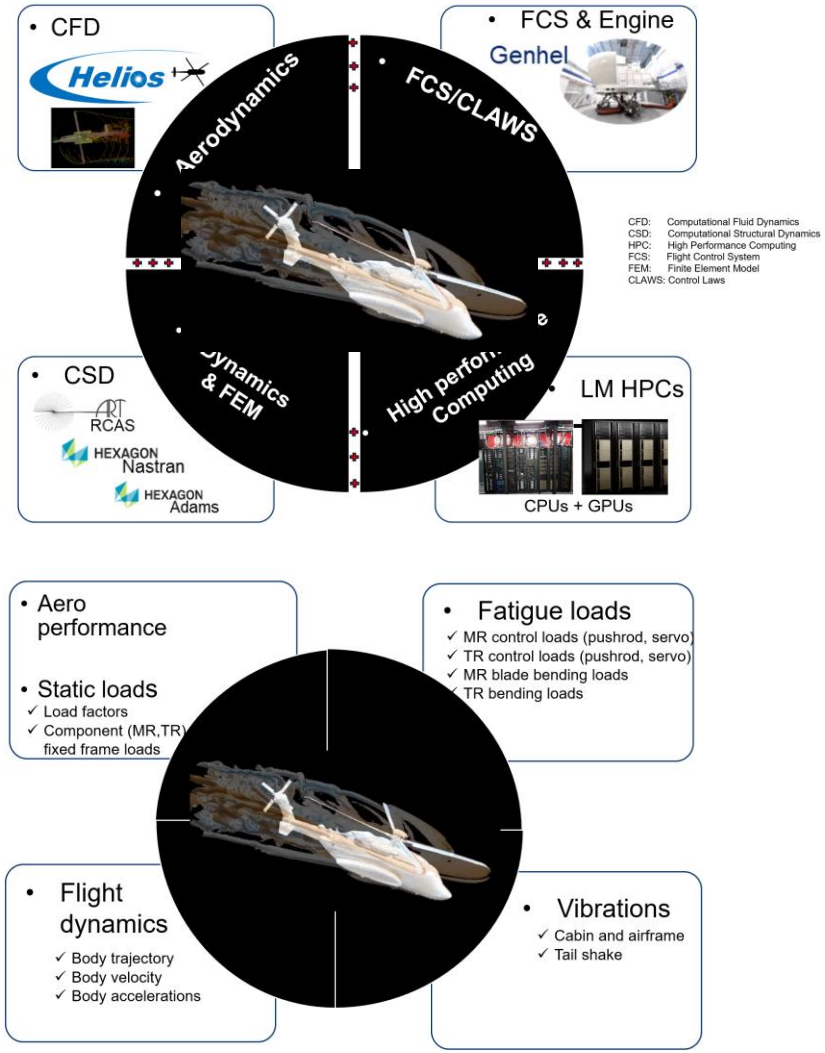
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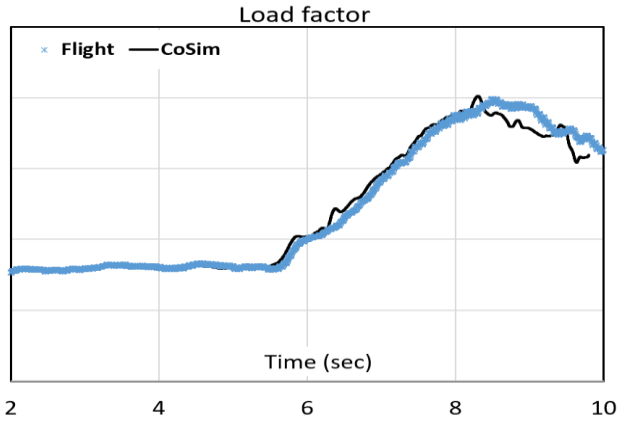
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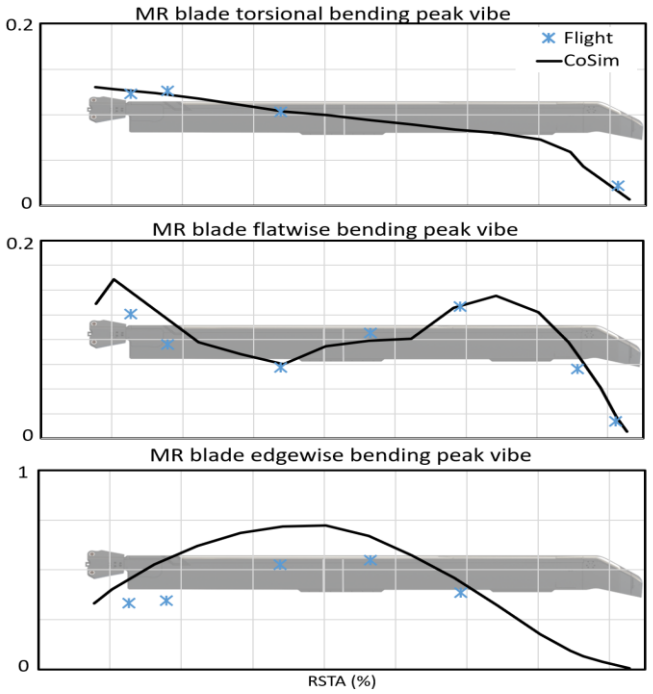
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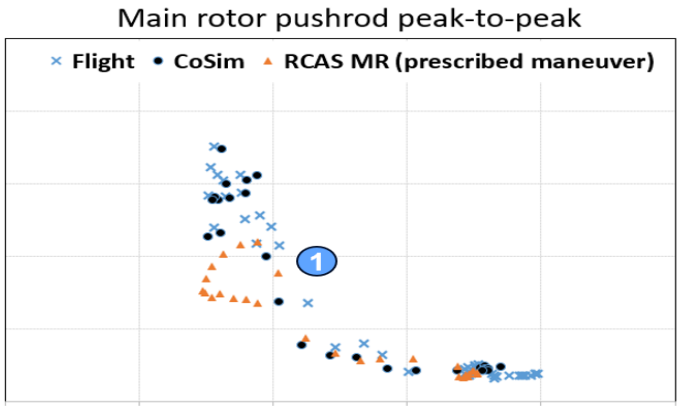
Additional CoSim Application Example: S-92A™ Helicopter

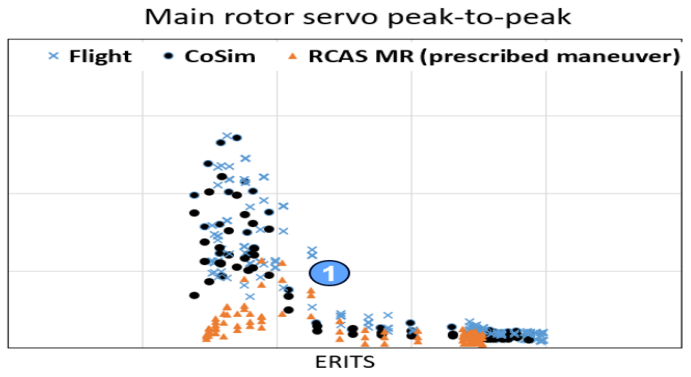
CoSim =

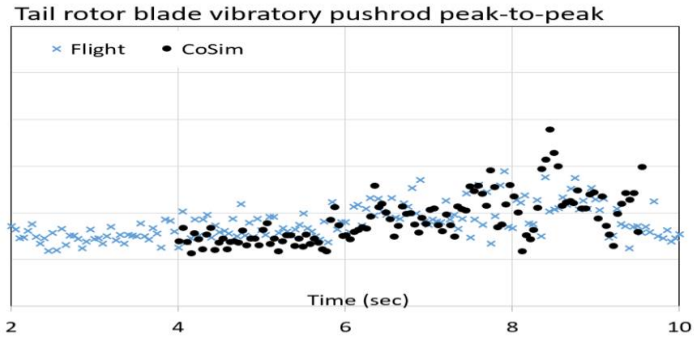












References:
Zhao, J., et al, "CFD-CSD-FCS Free-Flight Co-Simulation Correlations with S-92A Maneuvering Flight Tests," VFS 82nd Forum, May 2026.

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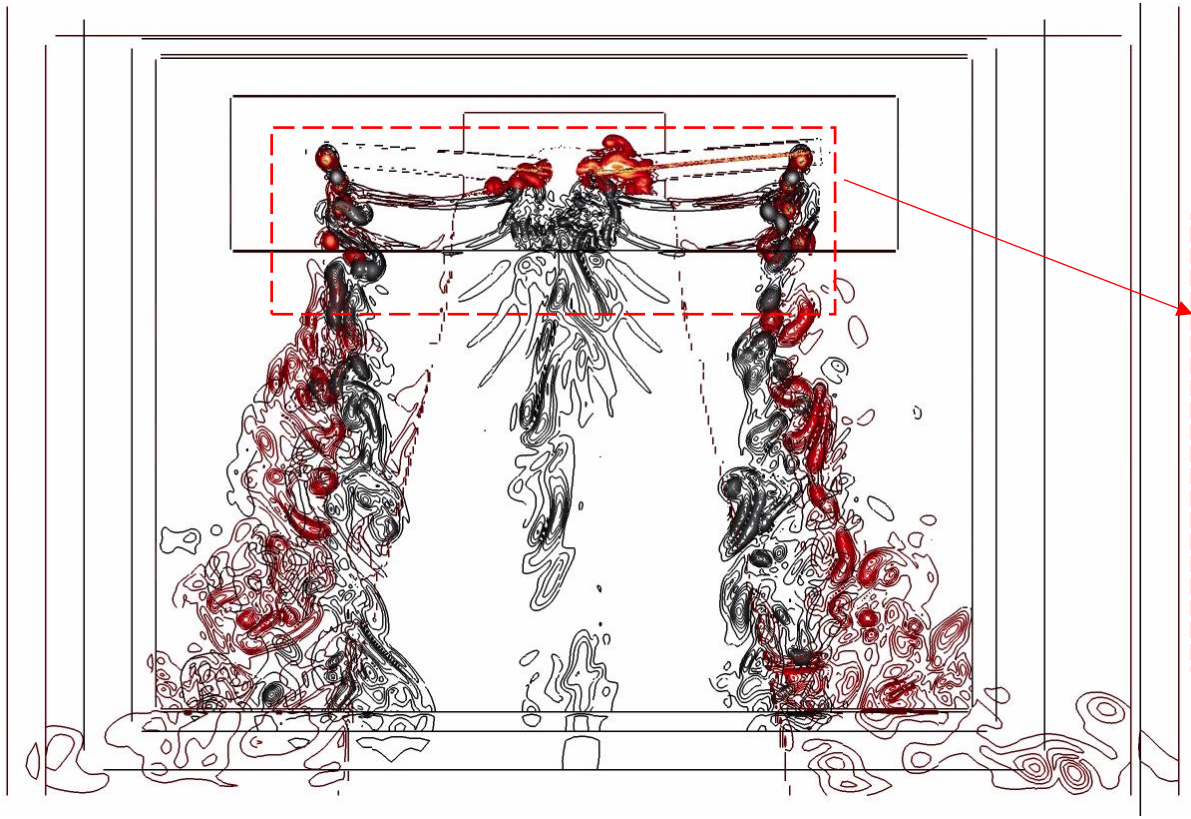
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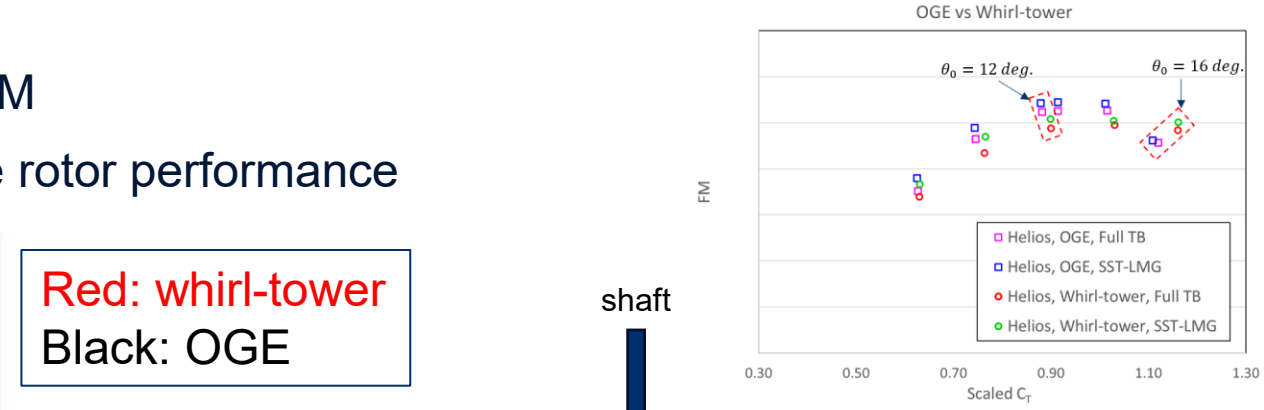
Flight Science Study

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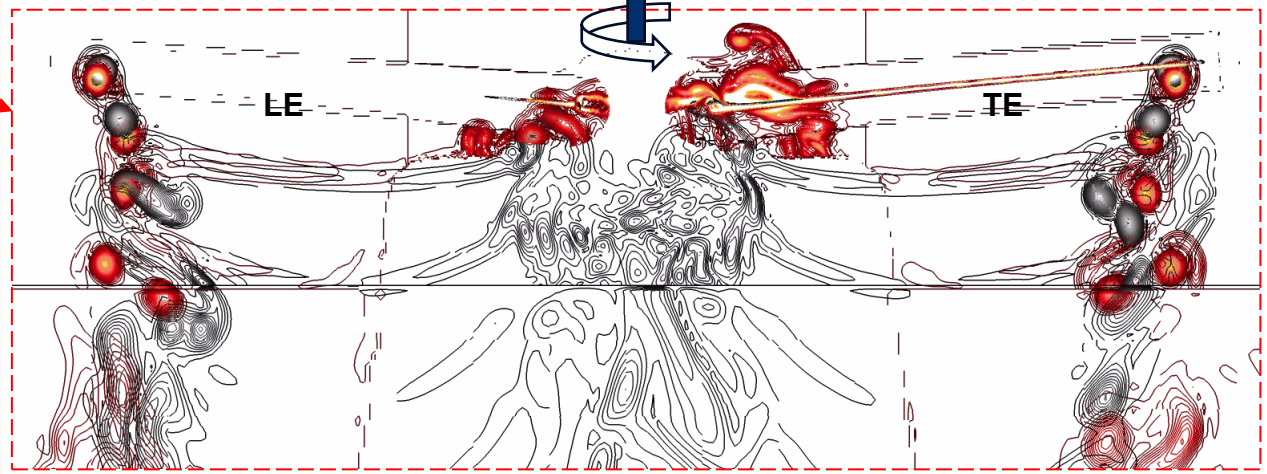
Whirl-Tower vs Isolated OGE: Tip-Vortex Trajectory

- Whirl-tower tip vortices pushed out more due to tower blockage→ stronger pull-down early age tip vortex→ weaker BVI→ Lower upwash and tip loading to swept tip→ less nose-down pitching moment→ less elastic twist down→ higher section pitch angle
- Weaker BVI reduces upwash benefit and reduces FM
- Whirl-tower test not necessarily represents accurate rotor performance





Red: whirl-tower
Black: OGE



References:
Min, B.-Y. and Wake, B. E., "Hovering Rotor Performance Prediction Throughout the Design Cycle," VFS 82nd Forum, May, 2026.

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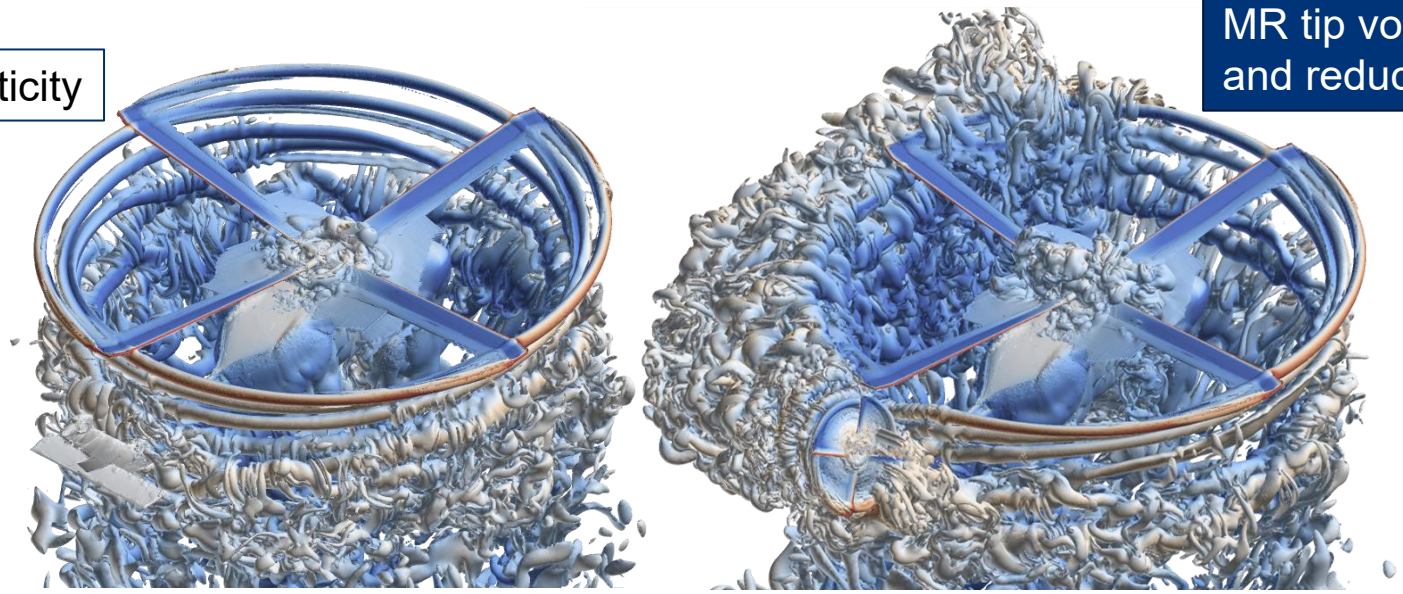
TR Impact to MR Performance

Without TR

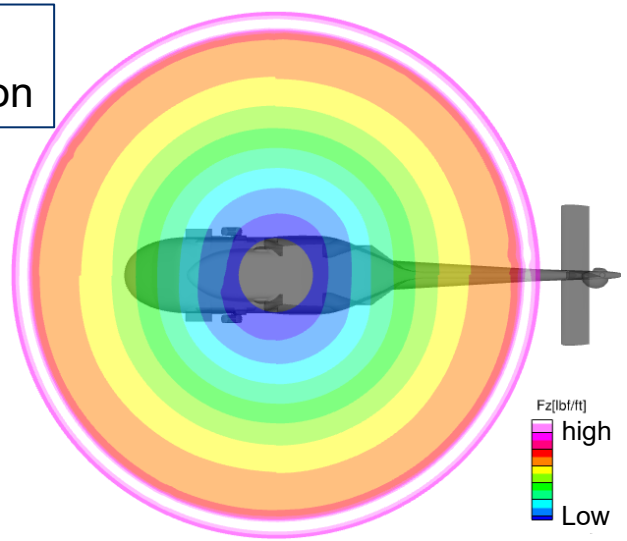
With TR

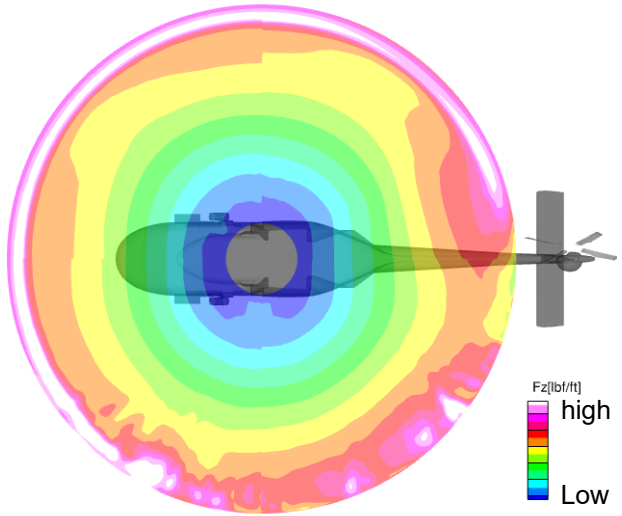
TR vortices disrupt MR tip vortex system and reduce tip loading

Iso-Vorticity

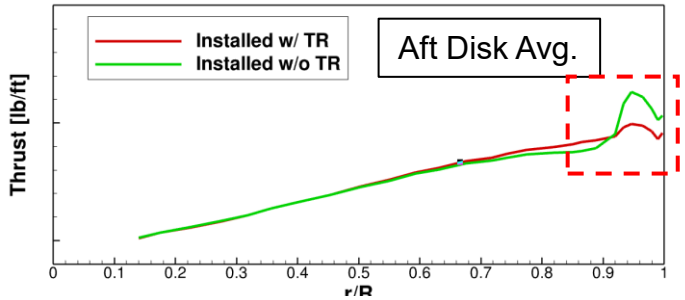


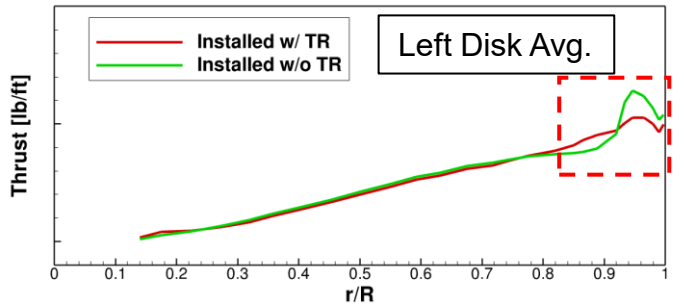
Thrust Distribution

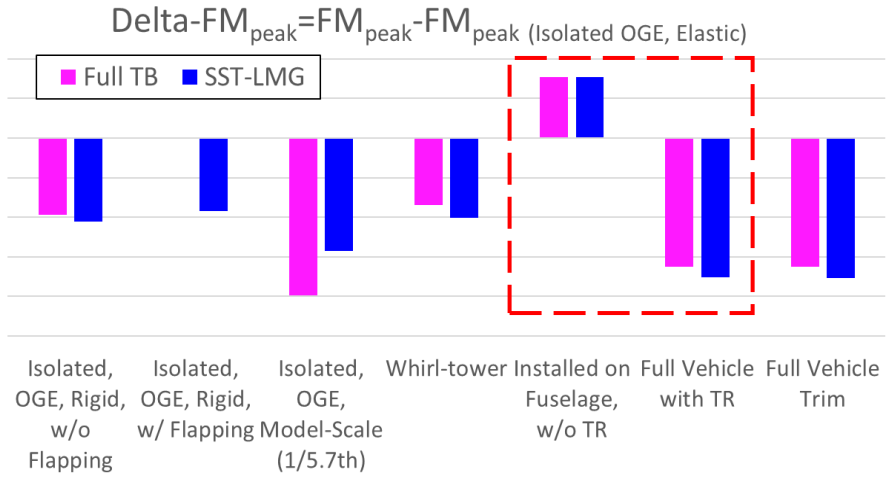




Averaged Section Thrust







References:
Min, B.-Y. and Wake, B. E., “Hovering Rotor Performance Prediction Throughout the Design Cycle,” VFS 82nd Forum, May, 2026.

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Applications in Development

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Other Aircraft Applications in Development

Helios in Use for Other Rotorcraft Configurations and Pending Applications

NOMAD™ Rotor Blown Wing





Wind-Tunnel Test and Flight-Test Validation On-Going

HEX 2-Rotor Tiltwing



HELIOS Applications Underway

S-70UAS U-Hawk™



Future Applications Anticipated

SIK2025050759

References:
Regan, M., Klimchenko, V., Sargent, D.C., et al, “Design and Development of Sikorsky's Rotor Blown Wing Demonstrator,” VFS 81st Forum, May 2025.

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Concluding Remarks

- Overset mesh is a ‘must-have’ technology for complex rotorcraft simulations
- Helios has been used in many applications in Sikorsky, leading to successful programs with digital-twin approach
  - Sikorsky has put lots of efforts to validate and improve our modeling approach for increased confidence
- For industrial product design purpose, fast & robust (and easy to setup) overset technique is necessary during early design stage (even with slight compromise in accuracy)
  - Fast large design space exploration

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